

Prophylactically Limiting Phantom Limb Pain using Collaborative Virtual Reality

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Figure 1: (Left)VR View of the patient (Right)Patient and family member using the system

ABSTRACT

Phantom limb pain (PLP), experienced by up to 80% of amputees, significantly impacts quality of life and often leads to opioid dependence. This paper introduces a novel collaborative virtual reality (VR) system, Healing and Empowerment After Limb-loss in XR (HEAL-XR), designed to prophylactically address PLP immediately following amputation surgery. Unlike existing VR therapies, HEAL-XR is tailored for early intervention, adapts to diverse amputation types and limited post-operative mobility, and features a collaborative mode enabling family members or clinicians to assist patients within the virtual environment. Through a 10-week longitudinal case study of a quadruple amputee who began using the system on post-operative day one, we demonstrate HEAL-XR's feasibility and potential efficacy. Future work will involve larger clinical trials, incorporating graded motor imagery, and refining controller attachment methods for diverse amputation scenarios.

Index Terms: Virtual Reality, Medical, Pain, Collaboration

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1 INTRODUCTION

Limb amputations are widespread throughout the world, with approximately 185,000 surgeries performed annually in the United States alone [12]. Immediately after the operation, many patients must endure tremendous pain. This includes residual limb pain at the surgical site, but most amputees also experience a phenomenon known as phantom limb pain (PLP) localized where their amputated limb once was [10]. Often, this leads to long-term use of opioids among amputees, raising concerns about the adverse effects of physical and psychiatric health [8]. Given that one of the great success stories of virtual reality (VR) has been pain reduction [4], a VR application specifically tailored for amputees could also be valuable, helping to alleviate acute and chronic phantom-limb pain, reducing dependence on opioids and leading to a higher quality of life.

This paper discusses the design of such a VR application for use by amputees immediately after surgery and during ongoing therapies, with the intention of using the VR system as a prophylactic therapy for phantom limb pain, that is, addressing the problem in the early stages or before it starts. In addition to peripheral (i.e., in the limb) interventions such as targeted muscle reinnervation and the regenerative peripheral nerve interface [6, 2], we hypothesized that the implementation of VR therapy postoperatively would be feasible and effective for acute PLP and could potentially reduce chronic PLP by influencing the cortical reorganization process from which PLP originates [5]. While use of VR therapy to mitigate PLP can be found in the literature [11], those identified had significant limitations with regards to inpatient clinical use, especially their suitability for some amputations and for patients immediately following their amputation procedure.

Amputations vary greatly, ranging from partial hand amputations to quadruple amputations of all four limbs. Following such medical procedures, patients may have limited mobility and ability to operate a VR device, despite the potential benefits. For example, a patient may not be able to move their arms, legs or head for safety or due to their condition. Additionally, the posture of the patient may be restrictive (e.g., lying down). As a result, asking patients to wear elaborate equipment or to perform calibration procedures and tasks may not be possible.

Despite these challenges, an overarching goal of the application was to create an active, engaging VR experience for the patient to maximize distraction from the painful real world experience. Moreover, for these patients, the experience of seeing an amputated limb may be both emotionally and physically painful, and so we had an additional goal to be able to help the patient visualize their avatar's body in any way their mind felt it should be, while being able to afford direct avatar control where possible.

In the following sections, we describe our application's design, which centers around the idea that patients often have others in their environment that can support their use of VR devices, especially if they are wearing one themselves. We further provide evidence of its effectiveness in the form of a longitudinal clinical case study of a trial patient who presented with a particularly challenging case of quadruple amputation and used the VR application for a period of 10 weeks, including postoperative day one. We conclude by describing the evolution of this system and needs going forward to address ongoing challenges to deployment.

2 RELATED WORK

Virtual Reality treatments have been studied and successfully applied in rehabilitation settings involving not only amputees but also stroke patients. VR-based mirror therapy systems such as "Augmented Reflection Technology in Virtual Reality" (ART-VR) have demonstrated feasibility in terms of patient acceptance and improvements in upper-limb functionality [3]. The reported effects from the ART-VR sessions, which lasted only one week, suggest that virtual therapy systems are not only accepted but also show potential for reducing pain and enhancing sensory feedback. The case study using HEAL-XR addresses this further through a 10-week-long intervention, providing extended exposure. This extended duration is expected to be critical for influencing cortical reorganization and reducing the risk of chronic PLP, highlighting the importance of sustained VR therapy in postoperative care for amputees.

Another study centered around virtual mirror therapy for PLP rehabilitation was conducted on a group of 19 patients (13 amputees, 6 brachial plexus avulsion injury patients) [7]. The system used an Oculus Rift VR headset in combination with a Leap Motion controller. The objective of the study was to explore VR rehabilitation's impact on phantom limb pain and phantom limb movement by introducing a virtual mirror therapy interactive activity that mirrored the patients' intact arm in place of their amputated (or paralyzed) arm. While the study demonstrated positive results in PLP pain reduction, it was limited by factors such as the novelty of headset use and the lack of long-term evaluation. In contrast, our case study not only tackled accessibility challenges, particularly for quadruple amputees, but also extended the intervention period to 10 weeks, allowing for a more comprehensive understanding of the long-term effects of VR rehabilitation. Moreover, the study did not explore the potential benefits of virtual reality for preventative therapy whereas HEAL-XR was applied shortly after amputation.

Due to the difficulty of controlling VR interfaces with one's legs, current lower-body VR solutions often avoid the challenge of lower-limb embodiment or rely on procedural animations to simulate leg movement, limiting their applicability for amputees. A VR-based physical therapy system has shown promise for improving accessibility in treating lower-body extremity injuries by offering



Figure 2: In application view of a user viewing a streamed webpage and the mirror.

innovative rehabilitation solutions [9]. This approach uses a remote therapy setup to guide patients through lower-body movements and poses in a gamified environment. The system incorporates the HTC Vive headset and its companion device, the "VIVE Tracker," which functions as a tracking puck and provides real-time feedback via direct line-of-sight with external base stations. Results from a 6-week in-clinic study demonstrated that this setup is both feasible and highly acceptable for patients with lower-extremity injuries.

HEAL-XR builds on this concept by introducing a collaborative multiplayer experience to enhance the benefits of remote therapy. This feature allows a second user, such as a clinician or family member, to guide the patient's limbs, addressing the challenges of lower-limb control and providing a "helping hand" in therapy. By enabling a more personalized and interactive approach, HEAL-XR improves accessibility for amputees and enables social engagement, ultimately contributing to better therapy outcomes.

3 SYSTEM

The VR application, Healing and Empowerment After Limb-loss in XR (HEAL-XR), was developed using Unity3D 2021.3.16, specifically targeting the Meta Quest 2 VR headsets. It was designed to run at 72 frames per second at all times in an untethered (no PC) configuration. While untethered operation limited the graphical fidelity of the system (e.g., no dynamic shadows), this choice provided increased portability and simplified clinical use. It also makes the system much more affordable, an important consideration for multi-headset configurations.

The virtual environment was a purchased Unity3D beach asset¹, which was further customized to reduce complexity to maintain performance. As seen in Figure 2, we also added a movable stereoscopic mirror to the beach scene to allow the patient to see their virtual bodies. Additionally, we added a video screen (see Figure 2) that could be used to watch television from within VR per a request from the patient in our case study. This was done using the Unity 3D WebRTC library², relayed through the pion selective forwarding unit³ on our server to broadcast streams from our website.

¹<https://assetstore.unity.com/packages/3d/environments/summer-beach-cartoon-pack-vr-mobile-17444>

²<https://github.com/Unity-Technologies/com.unity.webrtc>

³<https://github.com/ionorg/ion-sfu>

We used the Ready Player Me⁴ avatar system to provide unique full virtual bodies for each user. A web app⁵ was designed to facilitate this process. As the VR application starts, it transmits its hardware unique application identifier to our web server. At the same time a pairing code is displayed on the headset (See Figure 2). The user then goes to a website and enters the pairing code (See Figure 3). Following this, the user is taken to the Ready Player Me web app, and can configure their virtual body, choosing a Male or Female likeness and a variety of hair and clothing. This configuration results in a download link, which is then copied into the website, causing the VR application to download the avatar. This only needs to be done once for each user/headset combination, as the application retains the previous configuration.

Inverse kinematics (IK) was used to control the avatars. The Final IK package⁶ from the Unity3D asset store was used to facilitate this process, specifically the full-body VR IK components. To better support amputees, the end targets of the hands included customizable offsets that were in-turn moved by the tracked controllers. These offsets could be configured from within the application by grabbing them with their controller's hand trigger. This process allowed the controllers to be attached to whatever part of the limb remained to afford some control over the virtual limb. In all cases, the head was directly controlled by the headset tracking. The remaining posture was determined by manipulating targets for the feet and pelvis, allowing arbitrary configuration of the avatar pose.

To simplify initialization of this process, we configured several preset poses within the scene, attached to relevant virtual seats in the environment. The avatar could be rapidly moved between these postures by clicking a button on one of the controllers, after which more fine grained manipulation of the limbs could be performed by grabbing the targets and moving them directly using the motion controllers. The controllers could also be used to teleport to any arbitrary location using the thumbstick of either Quest 2 controller. Depending on the height above the ground, the avatar would either float or the feet would be placed at the location of raycasts originating from the shoulders and pointed downwards. The user could also adjust their position smoothly by pulling themselves in any direction using the index trigger on either Quest 2 controller. The preset offsets were saved per-headset between sessions so that the patient could easily revisit the scene.

Control over the hand was guided by two processes. Squeezing the hand triggers of the controllers would animate the hand into a fist. Alternatively, these actions could be controlled through an in-world menu, operated from either controller, including periodic animations of opening and closing, as amputees frequently reported PLP in the form of a tightly closed fist. When grasps intersected objects, they could be picked up, and fingers curl around objects using raycasts from the tips to the object centers.

As the above implies the ability to use the controllers, and amputees may lack not be able to fully do that, we designed the software to be multiuser⁷ and enabled anyone in the scene to control the IK targets, allowing the bodies to be moved around and controlled without requiring the patient to do so. This turned out to be one of the more challenging and fruitful design decisions. It enabled a family member or clinician to play an active role in the experience, near or far (facilitated by a public server). They could help the patient move between presets, or, as seen in Figure 4, then could even grab their virtual bodies (by the head) and move them directly through teleporting or pulling. During these movements the patient's headset screen was faded to black to mitigate the otherwise sickening effects of unintended motion. This aspect also makes the

⁴<https://readyplayer.me/>

⁵<https://github.com/velaboratory/vel-connect>

⁶<http://root-motion.com/#final-ik>

⁷<https://github.com/velaboratory/VelNetUnity>

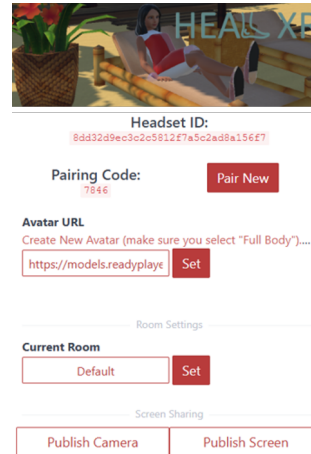


Figure 3: Web interface after entering a pairing code, allowing configuration of the network room, streaming, and avatar.



Figure 4: One user teleporting another by grabbing their head and teleporting themselves.

system highly versatile for a variety of amputee types. However, it also is the most time consuming aspect of system setup, requiring a few minutes of adjusting offsets with feedback from the patient. While these offsets are saved between sessions, if the attachment of the controllers changes, they must be readjusted. Also, as there is no tracking of the legs, the offsets of the legs need to be adjusted each time to match the patient's leg pose.

4 CASE STUDY

The first clinical test case of the VR application was a 52-year-old female admitted for quadruple amputation. Before this, she was healthy and active at baseline, an avid beach goer and surfer. She presented to the orthopedic surgery center as a transfer for bilateral below-knee, left trans-radial, and right partial-hand amputations. Amputations were performed during a single surgery and included regenerative peripheral nerve interfacing for her bilateral below-knee amputations and targeted muscle reinnervation for her left upper extremity trans-radial amputation. She was admitted to the intensive care unit for recovery. Within hours of surgery, she was fitted with stump protectors to her bilateral lower extremities and an immediate postoperative prosthesis to her left upper extremity. That evening, the patient began using the VR application with a Meta Quest 2 headset. A full-body Ready Player Me avatar was customized with her input depicting all four limbs (See Figure 1).

What remained of her upper extremities were tracked using a controller attached to her prosthesis (left upper extremity) and her

partial hand (right upper extremity), and the mapping was adapted to match this attachment from within the software. The patient and her family members were trained by the developers and clinicians in how to use the VR application together and what its capabilities were, but they were not given significant guidance on what to do within it, as this was the first real-world test of the user interface and was intended only to determine feasibility.

Though initially, this test case was intended for early conceptual feedback to guide future system design, the patient and family immediately responded well to the experience that we facilitated continued use of the application by leaving the VR headsets for their use. For the next 10 weeks (1 as an inpatient and the rest at rehab) she underwent self-directed VR therapy daily to treat acute PLP flares and to prophylactically decrease the development of chronic PLP by mitigating the central effects of cortical reorganization.

After graduating from rehab, she turned in her VR headsets and returned home (out of state). At 8 months post-op, she returned to the clinic for follow-up, where we were able to collect several surveys under an ongoing clinical trial to quantify her experiences, allowing us to better understand the efficacy, usability and feasibility of VR in the immediate post-operative setting and capture any lasting effects.

5 RESULTS

The patient reported experiencing phantom limb sensations in all four limbs upon awakening from anesthesia. She characterized these as pain, cramping and numbness. They were most noticeable along her left wrist and forearm and usually experienced as “numbness.” She rated her PLP as a 4/10 at baseline but as high as 9/10 frequently. She relied on VR therapy to get through these acute bouts of PLP and reported a decreased average PLP from 8/10 or 9/10 pre-session to an average of 2/10 during and for approximately 20 minutes after therapy. She found the VR sessions “soothed” and “calmed” and relieved “tightness” in her phantom limb. She rated the overall VR user-experience as 7/10, headset comfort as 8/10, and rated her willingness to recommend VR therapy to other amputees as a 10/10. She also noted that a highly effective technique was to “stroke” her phantom limb with her opposite limb, which could be done by herself or by others.

At 10 weeks post-op, she walked out of rehab on two below-the-knee prostheses and a left upper extremity body-powered prosthetic. At the time of this writing, she was using a myoelectric prosthesis for her left upper extremity. Her VR therapy ended at 10 weeks upon her return home, but she continued a multimodal pain regimen for residual limb pain and PLP. While she rated VR therapy equally effective to opioids, she expressed a strong preference for VR over opioids. She repeatedly expressed her desire to have continued VR therapy after discharge and rated her willingness to continue VR as 5/5.

Her System Usability Scale (SUS) score[1] for the VR system was 80, indicating excellent usability and acceptability. Her Simulator Sickness Questionnaire (SSQ) score was 0 indicating no discomfort, disorientation, nausea, or other symptoms of simulator sickness during use. The patient found the system highly learnable, consistent, and well-integrated, with minimal need for technical support or prior knowledge. The high SUS score and low SSQ support the feasibility of implementing VR therapy in the post-operative setting and suggests a low barrier to patient adoption and adherence with minimal side effects.

Eight months post-operatively, the patient reported an incidence of PLP that had decreased from daily to 3-4 times per week, with episodes lasting minutes rather than being constantly present. Pain intensity had also decreased to 4/10. The patient expressed a strong desire to continue using VR therapy, rating it as the most desired and effective treatment for her PLP (8/10), when compared to other forms of treatment as seen in Figure 5.

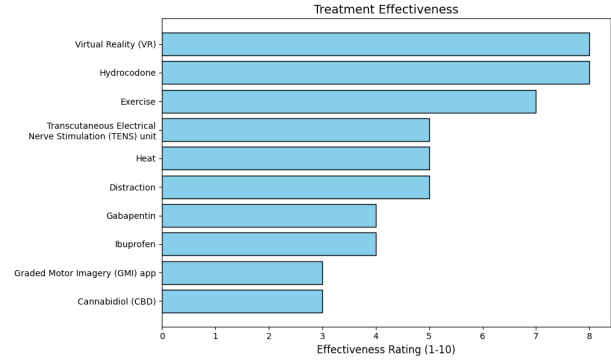


Figure 5: Patient self-reported pain relief effectiveness ratings.

6 DISCUSSION

As most research involving VR-based treatments for PLP have occurred in outpatient settings, we wanted to elucidate whether VR could be used safely and effectively in the inpatient setting. The patient’s initial positive experience with the system led to longitudinal use. Its reported learnability, consistency, and minimal side effects, suggests that VR therapy can be effectively incorporated into the acute clinical setting with minimal technical support, prior patient knowledge, or adverse effects. The system’s high usability with minimal risk of cybersickness may contribute to the patient’s willingness to continue using VR frequently.

The rapid and significant PLP reduction, from 8/9/10 to 2/10 during and after use, along with the patient’s strong preference for it over opioids, suggests that early VR may be an effective strategy for acute postoperative PLP that could possibly decrease opioid use. The patient’s continued experience of PLP at 8 months postoperatively, albeit at a reduced intensity and frequency, underscores the potential need for ongoing VR therapy to manage residual symptoms and prevent recurrence. The patient’s strong desire to continue using VR as a treatment for her PLP further supports the role of ongoing therapy in maintaining the benefits of early prophylactic intervention.

7 CONCLUSIONS AND FUTURE WORK

This system and case study demonstrated the promising role of VR therapy as a prophylactic intervention for acute postoperative PLP when initiated early during hospitalization following amputation. For patients that have undergone quadruple limb loss, the system may be especially powerful due to its collaboration features that address some usability concerns. However, the current system is not without limitations. It does not offer integrated guided therapies, such as graded motor imagery, which could improve access or address limited therapist time. In addition, the ergonomics of the residual limb tracking system could also be greatly improved, as it currently requires awkward attachment of the motion controllers to the patient. Future systems may have improved optical tracking systems that reduce the need for attached devices, though robust head-mounted full body tracking remains a challenging problem. Eye tracking and voice-control systems also have shown promise as interaction devices and could be used to afford further control the patient in HEAL-XR. As this study is the case of a single patient, obtained in retrospect, more feedback and detailed usage logs from patients are necessary to ensure that the system works for the majority of amputees, and reproducible clinical trials are required to provide medical guidance on its long term use and comparison with alternative treatments.

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